UNIT - 1

INTRODUCTION & PROPERTIES OF MATERIALS

History - Nanomaterials - Classification - Crystals - Band structure - Properties

1. History of Nano Science

Nanoscience is the study of phenomena and manipulation of matter at the atomic, molecular and macromolecular scale, where properties differ markedly from bulk. Nanotechnology is the application of this knowledge to design, produce and use useful devices.

Important milestones

Year	Contribution
~4th century AD	Lycurgus Cup (Rome): glass with Au-Ag nanoparticles looks green in reflected light, red in transmitted light
1857	Michael Faraday prepared colloidal gold (ruby red) and explained its colour is due to tiny particles
1905	Einstein estimated molecular size from sugar solution (about 1 nm)
1931	Ernst Ruska and Max Knoll invented the Transmission Electron Microscope (TEM)
1959	Richard Feynman lecture "There is plenty of room at the bottom" - idea of manipulating atoms
1974	Norio Taniguchi (Tokyo) first used the word "nano-technology"
1981	Binnig and Rohrer invented the Scanning Tunnelling Microscope (STM), Nobel Prize 1986
1985	Kroto, Curl and Smalley discovered fullerene C60 (Nobel 1996)
1986	Eric Drexler book "Engines of Creation"; Binnig, Quate, Gerber invented AFM

1991	Sumio Iijima discovered carbon nanotubes (CNTs) at NEC, Japan
2004	Geim and Novoselov isolated graphene using scotch tape (Nobel 2010)

Remember:
Feynman (1959) is called the "father of nanotechnology" for the vision; Taniguchi coined the term; Drexler popularised it.

2. Definitions

Nanometer

One nanometer is one billionth of a metre.

$$1\ nm = 10^{-9}\ m = 10\ Å = 10^{-3}\ \mu m$$

- Hydrogen atom diameter is about 0.1 nm; a DNA double helix is about 2 nm wide
- Human hair is about 80,000 nm thick; red blood cell about 7,000 nm
- Typical atom spacing in a solid is 0.2 – 0.3 nm, so 1 nm spans only 3 – 4 atoms

Nanomaterials

Materials having at least one dimension in the range 1 – 100 nm are called nanomaterials. Their size is comparable to the de Broglie wavelength / exciton Bohr radius of electrons, so quantum effects appear.

Nanotechnology

Design, characterisation, production and application of structures, devices and systems by controlling shape and size at the nanometre scale (1 – 100 nm).

Why is the nano scale special? (two main reasons)

1. **Surface to volume ratio increases** – for a sphere $S/V = 3/r$. Smaller r means a large fraction of atoms lie on the surface, so reactivity, catalysis, melting point etc. change.
2. **Quantum confinement** – when size is comparable to the electron

wavelength, energy levels become discrete and the band gap increases. Optical, electrical and magnetic behaviour changes.

Particle size	Total atoms	% atoms on surface
10 nm	30,000	~ 20 %
3 nm	4,000	~ 50 %
1 nm	30	~ 99 %

3. Classification of Nanomaterials

Based on the number of dimensions that are NOT confined to the nano range (i.e. free electron movement), nanomaterials are classified as 0D, 1D, 2D and 3D.

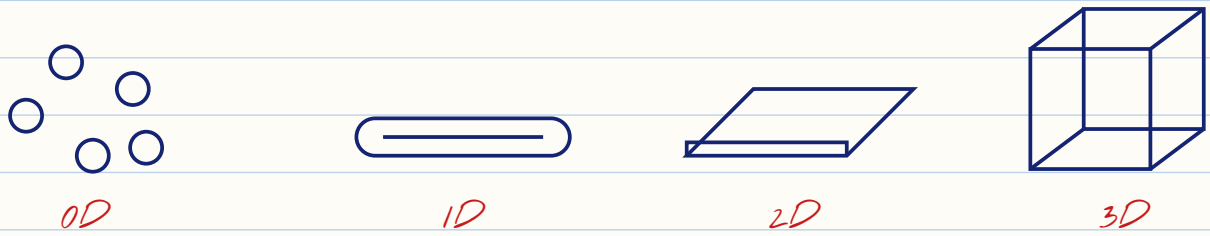


Fig: Dimensionality of nanomaterials (shape of the nano object)

Class	Confined in	Free in	Examples
0D	all 3 dimensions (x, y, z)	0 dimensions	Quantum dots, nanoparticles, fullerene C60, clusters
1D	2 dimensions	1 dimension (along length)	Nanowires, nanorods, nanotubes, nanofibres
2D	1 dimension (thickness)	2 dimensions	Graphene, thin films, quantum wells, nanosheets, MoS_2
3D	none (nano-structured bulk)	3 dimensions	Nanocrystalline bulk, nanocomposites, powders, multilayers, fullerite

Other ways of classifying

- **By material**: carbon-based (fullerene, CNT, graphene), metal-based (Au, Ag, Fe_3O_4), ceramic / metal oxide (TiO_2, ZnO), semiconductor (CdSe, CdS, Si), polymeric (dendrimers) and composite nanomaterials

- **By origin**: natural (volcanic dust, viruses, lotus leaf structure, gecko foot hairs), incidental (soot, welding fumes) and engineered (manufactured on purpose)
- **By structure (Gleiter)**: nanostructured materials are classified by shape (layered, rod, equiaxed) and chemical composition

4. Crystal Symmetries

A crystal = lattice + basis. A lattice is an infinite regular array of points; the basis is the atom or group of atoms attached to every lattice point. Unit cell: the smallest repeating block described by lattice parameters a, b, c and angles α, β, γ.

Symmetry operations

- **Translation** – shifting by a lattice vector leaves the crystal unchanged
- **Rotation** – only 1, 2, 3, 4 and 6-fold axes are allowed in crystals (5-fold cannot fill space without gaps)
- **Reflection (mirror plane)** – m
- **Inversion** – every point (x,y,z) goes to (−x,−y,−z), centre of symmetry
- **Rotation-inversion** and glide / screw operations (combined with translation) give the space groups

> **Fact:**
> 32 point groups and 230 space groups exist. Combining symmetry with lattice type gives 7 crystal systems and 14 Bravais lattices.

Seven crystal systems

System	Axes	Angles	Example
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90°$	NaCl, Cu, Si, Fe
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	TiO_2 (rutile), Sn
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	$BaSO_4$, Ga
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90°$, $\gamma = 120°$	Zn, Mg, graphite, ZnO

Trigonal (Rhombohedral)	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	Bi, calcite, quartz
Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90° \neq \beta$	Gypsum, monoclinic S
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	$CuSO_4.5H_2O$, $K_2Cr_2O_7$

14 Bravais lattices

- Cubic – 3: simple (SC), body-centred (BCC), face-centred (FCC)
- Tetragonal – 2: simple, body-centred
- Orthorhombic – 4: simple, base-centred, body-centred, face-centred
- Hexagonal – 1, Trigonal – 1
- Monoclinic – 2: simple, base-centred
- Triclinic – 1

Cubic cell	Atoms / cell	Coord. no.	Packing factor	Examples
SC	1	6	0.52	Polonium
BCC	2	8	0.68	Fe, Cr, W, Na
FCC	4	12	0.74	Cu, Al, Au, Ag
HCP	6	12	0.74	Mg, Zn, Ti

5. Crystal Directions and Planes

Crystal directions [u v w]

1. Choose the origin and draw the direction vector through it (or shift it to pass through the origin).
2. Take the coordinates of the point where the vector leaves the cell in units of a, b, c.
3. Multiply / divide by a common factor to get the smallest integers.
4. Enclose in square brackets [u v w]; negative index is written with a bar on top.

- x-axis = [100], y-axis = [010], z-axis = [001], body diagonal = [111], face diagonal = [110]

- Family of equivalent directions is written with angle brackets, e.g. ⟨100⟩ includes [100], [010], [001] and their negatives

Crystal planes – Miller indices (h K l)

Miller indices are three smallest integers that are the reciprocals of the intercepts of the plane on the three axes.

1. Find the intercepts of the plane on the x, y, z axes in terms of a, b, c (e.g. 2, 3, 1).
2. Take reciprocals (1/2, 1/3, 1).
3. Multiply by the LCM of the denominators (6): 3, 2, 6.
4. Enclose in round brackets: (3 2 6). A plane parallel to an axis has intercept ∞, so its index is 0.

> **Key:**
> Plane (hkl) cuts the axes at a/h, b/k, c/l. Family of planes is written {hkl}. In cubic crystals [hkl] is perpendicular to the plane (hkl).

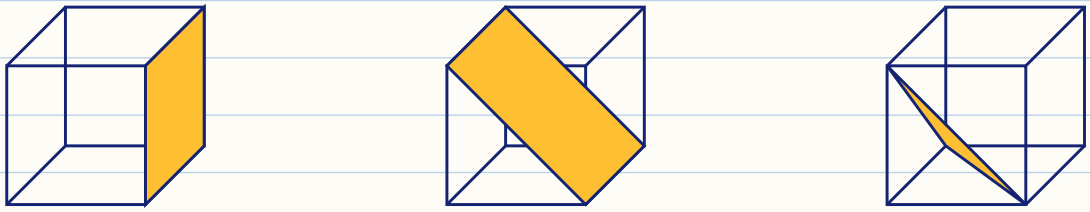


Fig: Important planes in a cubic lattice

Interplanar spacing

$$d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)} \qquad \text{(cubic)}$$

$$2 d \sin\theta = n \lambda \qquad \text{(Bragg's law)}$$

Example: for $a = 4$ Å, plane (111): $d = 4/\sqrt{3} = 2.31$ Å. For hexagonal: $1/d^2 = (4/3)(h^2 + hk + k^2)/a^2 + l^2/c^2$.

Hexagonal systems use four-index Miller-Bravais notation (h k i l) with $i = -(h + k)$.

Reciprocal lattice (needed for band structure)

- Reciprocal lattice vectors: $a^* = 2\pi (b \times c) / V$, etc. so that $a \cdot a^* = 2\pi$

- Every set of planes (hkl) in real space = one point in reciprocal space at distance $2\pi/d_{hkl}$
- **Brillouin zone** = Wigner-Seitz cell of the reciprocal lattice. First BZ of FCC is a truncated octahedron; of BCC is a rhombic dodecahedron. Symmetry points: Γ (centre), X, L, K.

6. Band Structure of Solids

Origin of bands

- An isolated atom has sharp energy levels. When N atoms are brought close together, wave functions overlap and, by Pauli exclusion, each level splits into N closely spaced levels
- These form a *quasi-continuous band*. Between bands lie forbidden gaps (E_g)
- Valence band (VB) = highest filled band. Conduction band (CB) = next empty / partly filled band

Bloch theorem

In a periodic potential $V(r + R) = V(r)$, solutions of the Schrodinger equation have the form:

$$\psi_K(r) = u_K(r)\, e^{i K \cdot r} \qquad \text{with } u(r + R) = u(r)$$

- K is the crystal wave vector; E(K) is periodic in K with period $2\pi/a$, so we only plot the first Brillouin zone ($-\pi/a$ to $+\pi/a$)
- **Kronig-Penney model** (rectangular periodic wells) gives: $\cos(Ka) = P \sin(\alpha a)/(\alpha a) + \cos(\alpha a)$. Allowed values are those where RHS lies between −1 and +1; others give forbidden gaps
- Near a band edge the effective mass $m^* = \hbar^2 / (d^2E/dK^2)$ replaces m: narrow band means heavy electrons

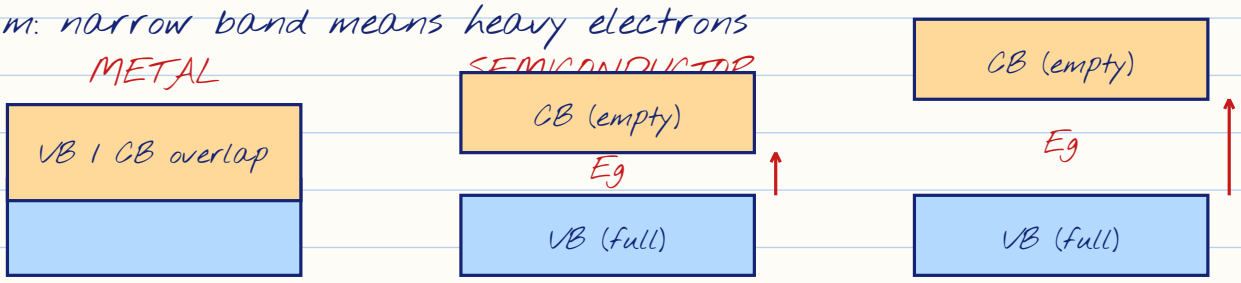


Fig: Band picture of metals, semiconductors and insulators

Property	Metal	Semiconductor	Insulator
Band gap Eg	0 (overlap / half filled)	0.1 – 3 eV (Si 1.12, Ge 0.67, GaAs 1.42)	> 3 eV (diamond 5.5, SiO2 ~9)
Resistivity (ohm m)	10^{-8}	10^{-5} to 10^{6}	10^{10} to 10^{20}
Temp. effect	Resistance rises with T	Resistance falls with T	Breakdown at high T

Direct and indirect band gap

- **Direct**: CB minimum and VB maximum at the same K (GaAs, InP, CdSe, ZnO). A photon alone can excite / recombine an electron so it is strongly emitting – used in LEDs and lasers
- **Indirect**: minimum and maximum at different K (Si, Ge). A phonon must help to conserve momentum, so emission is weak

$$E_F: \quad f(E) = 1 / [1 + \exp((E - E_F)/K_B T)]$$

Fermi level E_F lies inside the band for metals and near mid-gap for intrinsic semiconductors; n-type moves it towards the CB, p-type towards the VB.

7. Properties of Materials

For each property we study the bulk behaviour first and then how it changes when size drops to the nanometre scale.

7.1 Mechanical properties

- Defined by stress-strain behaviour: Young's modulus Y = stress/strain, hardness, yield strength, toughness, ductility
- Hardness and yield strength of nanocrystalline metals follow the Hall-Petch relation:

$$\sigma_y = \sigma_0 + K\, d^{-1/2}$$

- where d = grain size. Smaller grains mean more grain boundaries, which block dislocation motion, so the material is harder (Cu with 10 nm grains is about 5x harder than coarse Cu)
- Below d ≈ 10 – 20 nm the trend reverses (inverse Hall-Petch)

because grain-boundary sliding dominates

- Nano-ceramics become ductile / superplastic; CNTs have Young's modulus ≈ 1 TPa and tensile strength 50 – 100 GPa (about 100 times stronger than steel at 1/6 weight)
- Nanocomposites (clay-polymer, CNT-polymer) show large increase in strength with a very small filler fraction

7.2 Electrical properties

- Conductivity $\sigma = n e \mu = n e^2 \tau / m^*$. In bulk metals, resistivity comes from scattering by phonons, impurities and grain boundaries (Matthiessen's rule $\rho = \rho_T + \rho_0$)
- When the size becomes smaller than the electron **mean free path** (≈ 40 nm in Cu), surface and grain-boundary scattering raise the resistivity
- In very small conductors transport is ballistic; conductance is quantised in steps of $G_0 = 2e^2/h = 77.5\ \mu S$ (resistance 12.9 KΩ per channel)
- **Coulomb blockade**: charging energy $E_C = e^2/2C$ of a tiny island exceeds $K_B T$, so electrons tunnel one at a time (basis of the single-electron transistor)
- Carbon nanotubes can be metallic or semiconducting depending on chirality (n, m): metallic if (n – m) is a multiple of 3
- Semiconductor nanocrystals: band gap widens, carrier concentration changes, conductivity can be tuned by doping and size

7.3 Dielectric properties

- A dielectric is an insulator that polarises in an electric field. Polarisation $P = \varepsilon_0 \chi E$ and relative permittivity $\varepsilon_r = 1 + \chi$
- Mechanisms: electronic, ionic (atomic), orientational (dipolar) and space-charge (interfacial) polarisation. Each acts up to its own frequency limit
- Nano-structured dielectrics have a very large interface (grain-boundary) area, so **space-charge polarisation** rises and ε_r can become very high at low frequency (e.g. nano $BaTiO_3$)

$CaCu_3Ti_4O_{12}$)

- Below a critical size (~ 10 – 20 nm for $BaTiO_3$) ferroelectricity weakens and the cubic (paraelectric) phase is stabilised
- Nanocomposites of polymer + ceramic have high breakdown strength, so they are used in energy-storage capacitors, embedded capacitors and gate dielectrics (high-K: HfO_2)
- Dielectric loss tan $\delta = \varepsilon''/\varepsilon'$ depends on frequency; interfaces can raise loss

7.4 Thermal properties

- Lattice thermal conductivity $\kappa = (1/3)\, C\, v\, \lambda_{ph}$ (C = heat capacity, v = sound velocity, λ_{ph} = phonon mean free path)
- **Melting point falls** with size because surface atoms have fewer bonds: Au melts at 1064 °C in bulk but about 300 °C for 2 nm particles (Gibbs-Thomson effect)
- Phonon scattering at grain boundaries / surfaces reduces κ in nanostructures (Si nanowires have ~ 100 times smaller κ than bulk Si). This is great for thermoelectrics with figure of merit $ZT = S^2\sigma T/\kappa$
- Specific heat increases (more surface and grain-boundary atoms with extra vibrational modes); thermal expansion of nanocrystals is larger
- On the other hand, CNTs and graphene have extremely high thermal conductivity (3000 – 5000 W/mK), useful for heat spreaders
- Nanofluids (Al_2O_3, CuO or CNT in water) enhance heat transfer coefficients

7.5 Magnetic properties

Type	Behaviour	Susceptibility χ	Example
Diamagnetic	Weakly repelled, no permanent moment	small negative	Cu, Bi, water
Paramagnetic	Weakly attracted, random moments	small positive (Curie law)	Al, Pt, O_2

Ferromagnetic	Parallel aligned moments, domains, hysteresis, Curie temp T_C	large positive	Fe, Co, Ni
Antiferromagne	Neighbouring moments antiparallel	small positive; Neel temp T_N	MnO, Cr
Ferrimagnetic	Unequal antiparallel moments	large positive	Fe_3O_4, ferrites

- Ferromagnet breaks into domains to reduce magnetostatic energy. Below a critical single-domain size (~ 15 nm for Fe, 70 nm for Co) a particle becomes a single domain, so coercivity H_c is maximum
- Below that size thermal energy flips the moment: the particle is **superparamagnetic** (large moment, zero coercivity and remanence). Blocking temperature $T_B = KV / (25\ K_B)$ - where K = anisotropy constant, V = volume
- Surface atoms have broken exchange bonds, so saturation magnetisation often falls for very small particles. Antiferromagnetic nanoparticles can show weak ferromagnetism
- **Giant Magnetoresistance (GMR)**: multilayers of Fe/Cr or Co/Cu change resistance strongly with field (Fert and Gruenberg, Nobel 2007) - used in hard-disk read heads
- Applications: ferrofluids, MRI contrast agents, magnetic hyperthermia, high-density data storage, drug targeting

7.6 Opto-electronic properties

- Optoelectronics deals with devices that convert light into electricity or electricity into light: solar cells, LEDs, lasers, photodetectors
- **Quantum size effect**: as semiconductor size drops, the band gap widens, so absorption and emission shift to shorter wavelength (blue shift). CdSe dots change colour from red (large, ~ 6 nm) to blue (small, ~ 2 nm)
- **Surface Plasmon Resonance (SPR)**: collective oscillation of conduction electrons of Au / Ag nanoparticles in resonance with light. Au (~ 20 nm) absorbs near 520 nm, hence ruby-red colour; position shifts with size, shape and medium

- Exciton: electron-hole pair bound by Coulomb force with Bohr radius a_B. When particle radius < a_B the strong confinement regime occurs; exciton binding energy and oscillator strength rise
- Photoluminescence quantum yield of dots can exceed 90 %; sharp, size-tunable emission; large absorption cross-section
- Bulk Si is indirect and non-emitting, but porous / nano Si emits visible light

8. Effect of Size Reduction on Properties

Property	Change as size decreases
Surface area / volume	Increases enormously (SIV = 3/r); more reactive, better catalysts
Melting point	Decreases (Au 1064 °C to ~ 300 °C at 2 nm)
Lattice parameter	Usually contracts slightly due to surface stress
Hardness / strength	Increases (Hall-Petch) until about 10 nm
Band gap	Increases for semiconductors (blue shift)
Electrical resistivity	Increases for metals (surface scattering); conductance quantised
Magnetism	Single domain, then superparamagnetism; coercivity peaks
Optical	SPR colour in metals, size-dependent fluorescence in dots
Thermal conductivity	Decreases (phonon scattering); good for thermoelectrics
Dielectric	Space-charge polarisation increases; ferroelectricity vanishes under critical size

Summary:
Two root causes: (1) surface / interface atoms dominate; (2) quantum confinement of electrons, phonons and excitons.

9. Electronic Structure of Nanomaterials

Quantum confinement

If a dimension L is comparable with the de Broglie wavelength $\lambda = h / \sqrt{(2m^*E)}$ of the carriers, the electron is confined and its energy is quantised. For a particle in a 1D infinite well:

$$E_n = n^2 h^2 / (8 m^* L^2), \qquad n = 1, 2, 3 \ldots$$

- Energy $\propto 1/L^2$. Smaller L means bigger level spacing and a larger gap
- **Quantum well** (2D system): confined in 1 direction, free in 2. **Quantum wire** (1D): confined in 2, free in 1. **Quantum dot** (0D): confined in all 3 - "artificial atom"

Density of states g(E)

System	Free dim.	g(E) dependence	Shape
Bulk (3D)	3	$\propto E^{1/2}$	Smooth parabola
Quantum well (2D)	2	independent of E in each sub-band	Staircase
Quantum wire (1D)	1	$\propto E^{-1/2}$	Sharp spikes (van Hove singularities)
Quantum dot (0D)	0	$\delta(E - E_n)$	Discrete delta lines like an atom

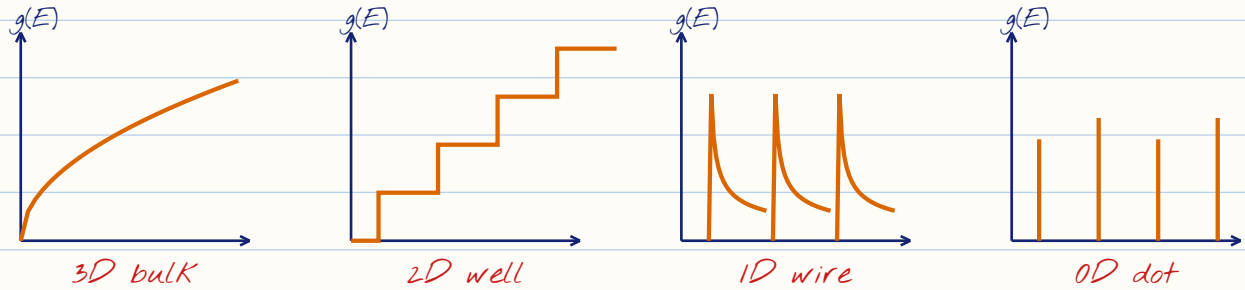


Fig: Density of states with decreasing dimensionality

Brus equation - size dependence of band gap

For a spherical semiconductor nanocrystal of radius R, the effective optical gap is:

$$E_g(R) = E_g(bulk) + \hbar^2\pi^2 / (2R^2)(1/m_e^* + 1/m_h^*) - 1.8\, e^2 / (4\pi\, \varepsilon_0\, \varepsilon_r\, R)$$

- 1st term: bulk band gap. 2nd term: confinement energy $\propto 1/R^2$ (blue shift). 3rd term: electron-hole Coulomb attraction $\propto 1/R$ (red shift, smaller effect)
- **Weak confinement**: $R > a_B$; **strong confinement**: $R < a_B$ (exciton Bohr radius $a_B = 4\pi\, \varepsilon_0\, \varepsilon_r\, \hbar^2 / (\mu\, e^2)$)
- Examples of a_B: CdS 3 nm, CdSe 5.6 nm, GaAs 12 nm, PbS 18 nm

Other consequences

- Discrete, atom-like levels give sharp absorption / emission peaks and "spectroscopic" size sorting
- Fermi level concept: for metal clusters, level spacing $\delta \approx 4E_F / 3N$ becomes larger than k_BT at small N: **Kubo gap**, so a metal cluster may become non-metallic below ~ 1 – 2 nm
- "Magic numbers" of atoms (e.g. 2, 8, 20, 40, 58 … for alkali clusters; 13, 55, 147 for icosahedral packing) indicate specially stable clusters, as in the jellium shell model

Quick Revision – Important Questions

1. Define nanometer, nanomaterial and nanotechnology. Why do properties change at the nano scale?
2. Classify nanomaterials with examples (0D, 1D, 2D, 3D).
3. Explain the seven crystal systems and 14 Bravais lattices.
4. How are directions and planes represented? Find the Miller indices of a plane with intercepts 2a, 3b, c.
5. Describe the band structure of metals, semiconductors and insulators; distinguish direct and indirect gaps.
6. Write a note on mechanical and magnetic properties of nanomaterials (Hall-Petch, superparamagnetism).
7. Explain electronic structure of nanomaterials; derive the density of states for 3D, 2D, 1D, 0D.
8. Explain the effect of size reduction on optical, electrical and thermal properties.